Notes on Bayesian Inference

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1. Likelihoods

What parameter makes our data most likely?

H H T T H H H H H T

If I flip this coin again, what is the probability it lands heads?

$$P(X = H) = ?$$

You might answer ½ — which is reasonable, but I never said it was a fair coin.

Better question: What is the most likely probability of heads that produced these flips?

Formalizing the Problem

Define a probability mass function:

$$P(X = H) = \theta, \quad \text{with } \theta \in [0, 1]$$
$$P(X = T) = 1 - \theta$$

A coin flip is a sample $X \sim P(X)$, where X is a random variable that can take the value H or T

Joint probability of the sequence HHTTHHHHHT:

$$\theta \cdot \theta \cdot (1-\theta) \cdot (1-\theta) \cdot \theta \cdot \theta \cdot \theta \cdot \theta \cdot \theta \cdot (1-\theta)$$

$$= \theta^7 (1 - \theta)^3$$

The Binomial Distribution

We want the probability of k heads in n flips (any order)

Count all arrangements of 7 heads in 10 flips:
HHHHHHHTTT, HHHHHHTHTT, HHHHHTHHTT, ...

The number of such arrangements is $C(n, k) = C(10, 7) = 120$

The Binomial distribution — n independent Bernoulli trials with k successes:

$$P(X = k \mid \theta) = C(n, k) \cdot \theta^k \cdot (1 - \theta)^{n-k}$$

The Likelihood Function

What parameter θ makes the data most likely?

Define the likelihood function:

$$L(\theta \mid D) = P(D \mid \theta)$$

L(θ | D) and P(D | θ) are the same quantity — we use L to highlight that θ is the variable (not the data D)

We want to find the θ that maximizes this — maximum likelihood estimation:

$$\max_{\theta} \; L(\theta \mid D) = C(n, k) \cdot \theta^{k}(1-\theta)^{n-k}$$

Maximum Likelihood Estimation

Apply log (monotonic — preserves maximum) → log-likelihood:

$$\ell(\theta \mid D) = \log C(n,k) + k \log \theta + (n-k) \log(1-\theta)$$

Take the partial derivative and set to zero:

$$\partial\ell/\partial\theta = k/\theta - (n-k)/(1-\theta) = 0$$

Solve for θ:

$$k(1-\theta) = (n-k)\theta \rightarrow k = n\theta$$

$$\theta^*\_{MLE} = k / n = 7 / 10 = 0.7$$

The MLE is just the empirical mean: successes / trials

2. Priors

Encoding beliefs before seeing data

Example 1: Flip a 1-franc coin 5 times

H H T H H → $\theta^*\_MLE = 4/5 = 0.8$

Example 2: Flip a 1-franc coin 10 times

H H H T H H H H H H → $\theta^*\_MLE = 9/10 = 0.9$

Does θ = 0.9 feel right for a real franc coin?

If 100 people each flipped 10 fair coins, roughly 1 would get 9 heads.
Can we encode the idea that we're not entirely convinced the coin is biased?

Frequentist

There is a "true" parameter θ that we don't have access to

We observe data until our estimators converge to that true value

Estimates are noisy with less data, but only samples can inform those estimates

Bayesian

We can incorporate prior knowledge or beliefs before seeing data

Data lets us update those beliefs

Focus of this lecture: encoding and updating beliefs about parameters

The Prior Density

To encode prior beliefs, we define a prior density $p(\theta)$

Notation: $P(\cdot)$ for discrete (mass functions)
$p(\cdot)$ for continuous (density functions)

We use the Beta distribution:

$$p(\theta) = [1 / B(\alpha, \beta)] \cdot \theta^{(\alpha-1)} \cdot (1-\theta)^{(\beta-1)}$$

$B(\alpha, \beta)$ is a normalization constant:

$$B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta) / \Gamma(\alpha+\beta) = \int_0^1 \theta^{(\alpha-1)}(1-\theta)^{(\beta-1)}\, d\theta$$

α and β control the shape — they encode your belief about θ

$\alpha = 1, \beta = 1$
(Uniform)

All values equally likely

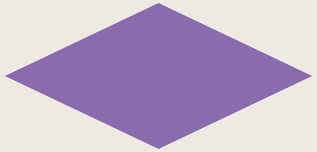
$\alpha = 5, \beta = 5$
(Symmetric)

Centered around 0.5

$\alpha = 2, \beta = 8$
(Left-skewed)

Believes θ is small

$\alpha = 20, \beta = 20$
(Strong prior)



Very confident $\theta \approx 0.5$

$\theta \rightarrow$ 0 1

The prior is a choice you are making — as in you, a person with subjective ideas about the world.

For a Swiss franc coin, we might set $\alpha = \beta \approx 20$ to encode strong confidence that θ is near 0.5.

3. Posteriors

Updating beliefs with observed data

Bayes' Rule

Recall Bayes' rule:

$$P(A \mid B) = P(B \mid A) \cdot P(A) \;/\; P(B)$$

Applied to our problem:

$$p(\theta \mid D) = p(D \mid \theta) \cdot p(\theta) \;/\; p(D)$$

Where $p(D) = \int p(D \mid \theta') \, p(\theta') \, d\theta'$ is the marginal likelihood

This denominator is generally intractable — we can't easily
integrate over all possible values of θ
(and it's effectively impossible when θ represents multiple parameters)

Since p(D) is fixed and doesn't depend on θ, the posterior is proportional to the numerator:

posterior ∝ likelihood × prior

$$p(\theta \mid D) \propto p(D \mid \theta) \cdot p(\theta)$$

$p(\theta \mid D)$

Posterior
What we believe about θ after seeing data

$p(D \mid \theta)$

Likelihood
How likely the data is for each θ

$p(\theta)$

Prior
What we believed about θ before seeing data

Deriving the Posterior

Substitute our Binomial likelihood and Beta prior:

$$p(\theta \mid D) \propto C(n,k)\, \theta^k(1-\theta)^{n-k} \cdot [1/B(\alpha,\beta)]\, \theta\text{^}(\alpha-1)(1-\theta)\text{^}(\beta-1)$$

Drop constants (they don't depend on θ):

$$\propto\ \theta^k(1-\theta)^{n-k} \cdot \theta\text{^}(\alpha-1)(1-\theta)\text{^}(\beta-1)$$

Combine exponents:

$$\propto\ \theta\text{^}(k+\alpha-1) \cdot (1-\theta)\text{^}(n-k+\beta-1)$$

This is exactly a $\text{Beta}(k + \alpha,\ n - k + \beta)$ distribution!

We can recover the normalizing constant because the posterior must integrate to 1: $Z = B(k+\alpha, n-k+\beta)$

Conjugate Prior & MAP Estimation

Conjugate prior: the posterior has the same functional form as the prior (given the likelihood). Here: Beta + Binomial → Beta

Maximum a posteriori (MAP) — the mode of the posterior:

$$\theta^*_{MAP} = (k + \alpha - 1) \; / \; (n + \alpha + \beta - 2)$$

The posterior mean (different from the mode in general):

$$E[\theta \mid D] = (k + \alpha) \; / \; (n + \alpha + \beta)$$

When the posterior is symmetric, mode and mean coincide.
In general they differ.

Key: as n grows large, both MAP and mean converge to the MLE k/n — the data overwhelms the prior.

Prior "pseudo-observations" (α = 5 heads, β = 5 tails)

H H H H H T T T T T

$\alpha = 5$ $\beta = 5$

Observed data (k = 7 heads, n − k = 3 tails)

H H H H H H H T T T

$k = 7$ heads, $n - k = 3$ tails

Prior

Beta(5, 5)
Centered at $\theta = 0.5$

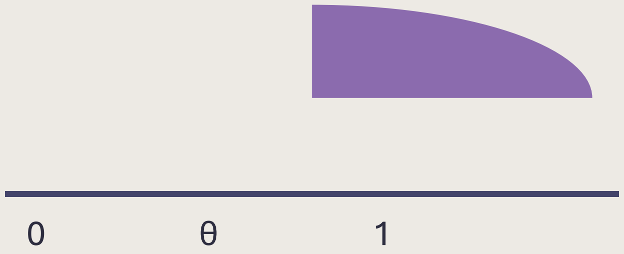
×

Likelihood

Binom(7 | 10, θ)
Peaks at $\theta = 0.7$

∝

Posterior



Beta(12, 8)
Shifted toward $\theta = 0.6$

Posterior: $\text{Beta}(k + \alpha,\ n - k + \beta) = \text{Beta}(7 + 5,\ 3 + 5) = \text{Beta}(12, 8)$

$\theta^*\_MAP = (7 + 5 - 1) / (10 + 5 + 5 - 2) = 11/18 \approx 0.61$ (vs. MLE = 0.70)

↑ *The prior pulls our estimate from 0.7 toward 0.5*

Interpreting α and β

The update rule reveals a beautiful interpretation:

$$\theta^*_{MAP} = (\text{\# observed heads} + \text{\# prior heads}) \,/\, (\text{\# observed flips} + \text{\# prior flips})$$

α acts like "imaginary heads you've already seen"
β acts like "imaginary tails you've already seen"
α + β = strength of your prior belief (total pseudo-observations)

As $n \to \infty$, both MAP and posterior mean → MLE (k/n)
The data eventually overwhelms any finite prior.

You don't even need the normalizing constant to get the MAP — just maximize the unnormalized quantity!

4. The Posterior Predictive

Making predictions under uncertainty

The Question Revisited

If I flip this coin again, what is $P(X = H \mid D)$?

MLE approach: plug in $\theta^* = k/n$
→ This ignores our uncertainty about θ

Bayesian approach: average over all possible θ values, weighted by the posterior:

$$P(X = H \mid D) = \int_0^1 P(X = H \mid \theta) \cdot p(\theta \mid D) \, d\theta$$

$$= \int_0^1 \theta \cdot p(\theta \mid D) \, d\theta$$

$$= E[\theta \mid D]$$

$$P(X = H \mid D) = E[\theta \mid D] = (k + \alpha) / (n + \alpha + \beta)$$

The posterior predictive equals the posterior mean — your best prediction for the next flip is your current average belief about θ.

Worked example: $k = 7$, $n = 10$, $\alpha = \beta = 5$

$P(X = H \mid D) = (7 + 5) / (10 + 5 + 5) = 12 / 20 = 0.60$

Compare: MLE = 7/10 = 0.70
The prior pulls the prediction toward 0.5, reflecting our belief that the coin is probably fair.

Consider: 1 head in 1 flip

H

MLE

$\theta^* = k/n = 1/1 = 1.0$

"The coin ALWAYS lands heads"

Would you bet the house on this?

Posterior Predictive

With Beta(2,2) prior:
$(1+2)/(1+2+2) = 3/5 = 0.60$

Shifted toward heads,
but not fanatically so

When we have limited data and real uncertainty about θ, averaging over that uncertainty gives better-calibrated predictions than committing to a single value.

1 Likelihood

$L(\theta|D) = P(D|\theta)$ measures how likely the data is for each θ. MLE maximizes this: $\theta^* = k/n$

2 Prior

$p(\theta)$ encodes beliefs before data. The Beta distribution parameterized by α, β is a flexible choice.

3 Posterior

posterior $\propto$ likelihood × prior. Beta prior + Binomial likelihood gives a Beta posterior (conjugacy).

4 Prediction

Average over posterior uncertainty for calibrated predictions: $P(H|D) = (k+\alpha)/(n+\alpha+\beta)$

The Bayesian framework: encode beliefs → observe data → update beliefs → make predictions